

Digital Design and Computer Organisation Laboratory

UE23CS251A

3rd Semester, Academic Year 2024-25

Date:12/09/2024

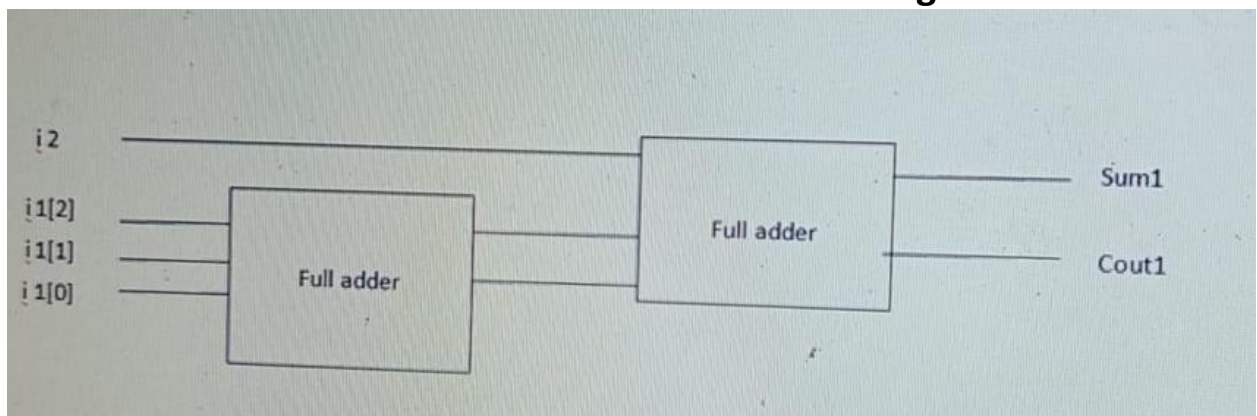
Name: Dipak Chaudhari	SRN: PES1UG24CS804	Section K
--------------------------	-----------------------	--------------

Week# 3

Program Number: Circuit 1

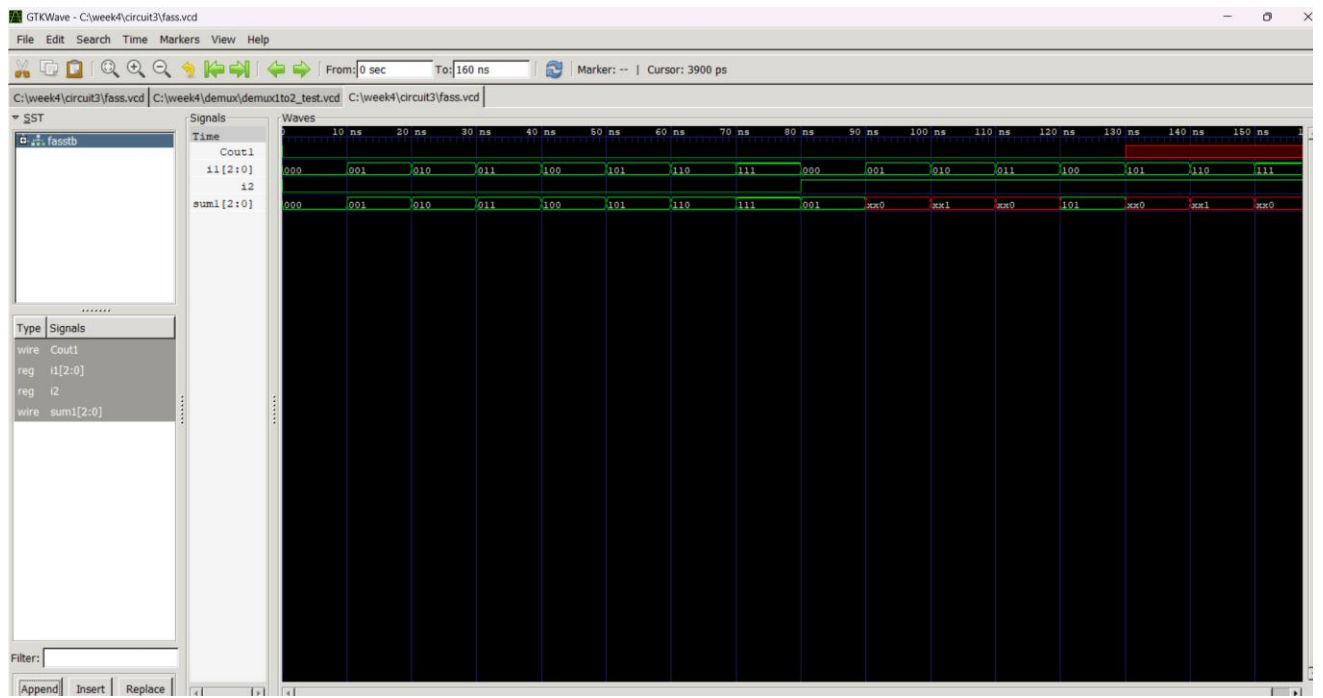
Title:

WRITE A VERILOG PROGRAM TO MODEL A Circuit given below



```
1  module fass(  
2      input wire [2:0] i1,  
3      input wire i2,  
4      output wire [2:0] sum1,  
5      output wire Cout1  
6  );  
7  
8      wire c;  
9      full_adder fa1 (i1[0], i2, 1'b0, sum1[0], c);  
10     full_adder fa2 (i1[1], 1'b0, c, sum1[1], c);  
11     full_adder fa3 (i1[2], 1'b0, c, sum1[2], Cout1);  
12 endmodule  
13  
14 module full_adder(  
15     input wire a,  
16     input wire b,  
17     input wire cin,  
18     output wire sum,  
19     output wire cout  
20 );  
21  
22     assign sum = a ^ b ^ cin;  
23     assign cout = (a & b) | (a & cin) | (b & cin);  
24 endmodule
```

```
≡ fass.v × ≡ fasstb.v ×
circuit3 > ≡ fasstb.v
1  `timescale 1ns / 1ps
2
3  module fasstb;
4
5      reg [2:0] i1;
6      reg i2;
7      wire [2:0] sum1;
8      wire Cout1;
9
10     fass dut (i1, i2, sum1, Cout1);
11
12     initial begin
13         $dumpfile("fass.vcd");
14         $dumpvars(0, fasstb);
15
16         // Test cases
17         i1 = 3'b000; i2 = 1'b0; #10;
18         i1 = 3'b001; i2 = 1'b0; #10;
19         i1 = 3'b010; i2 = 1'b0; #10;
20         i1 = 3'b011; i2 = 1'b0; #10;
21         i1 = 3'b100; i2 = 1'b0; #10;
22         i1 = 3'b101; i2 = 1'b0; #10;
23         i1 = 3'b110; i2 = 1'b0; #10;
24         i1 = 3'b111; i2 = 1'b0; #10;
25
26         i1 = 3'b000; i2 = 1'b1; #10;
27         i1 = 3'b001; i2 = 1'b1; #10;
28         i1 = 3'b010; i2 = 1'b1; #10;
29         i1 = 3'b011; i2 = 1'b1; #10;
30         i1 = 3'b100; i2 = 1'b1; #10;
31         i1 = 3'b101; i2 = 1'b1; #10;
32         i1 = 3'b110; i2 = 1'b1; #10;
33         i1 = 3'b111; i2 = 1'b1; #10;
34
35     $finish;
```



Digital Design and Computer Organisation Laboratory

UE23CS251A

3rd Semester, Academic Year 2024-25

Date:12/09/2024

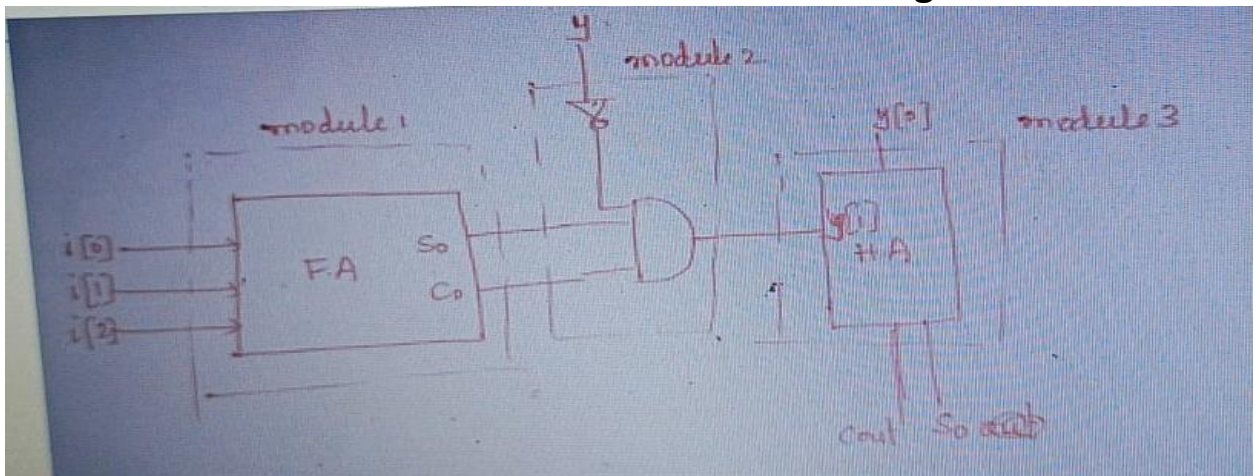
Name: Dipak Chaudhari	SRN: PES1UG24CS804	Section K
--------------------------	-----------------------	--------------

Week# 4

Program Number: Circuit 2

Title:

WRITE A VERILOG PROGRAM TO MODEL A Circuit given below



```

full_adder_vector.v  full_adder_vector_tb.v x
favector > full_adder_vector_tb.v
1  `timescale 1ns / 1ps
2
3  module full_adder_vector_tb;
4
5      reg [2:0] i1;
6      reg i2;
7      wire [1:0] sum1;
8      wire cout1;
9
10     full_adder_vector dut (
11         .i1(i1),
12         .i2(i2),
13         .sum1(sum1),
14         .cout1(cout1)
15     );
16
17     initial begin
18         i1 = 3'b000;
19         i2 = 1'b0;
20
21         #10;
22         $display("i1 = %b, i2 = %b, sum1 = %b, cout1 = %b", i1, i2, sum1, cout1);
23
24         i1 = 3'b001;
25         i2 = 1'b1;
26         #10;
27         $display("i1 = %b, i2 = %b, sum1 = %b, cout1 = %b", i1, i2, sum1, cout1);
28
29         i1 = 3'b011;
30         i2 = 1'b0;
31         #10;
32         $display("i1 = %b, i2 = %b, sum1 = %b, cout1 = %b", i1, i2, sum1, cout1);
33
34         i1 = 3'b111;
35         i2 = 1'b1;
36         #10;
37         $display("i1 = %b, i2 = %b, sum1 = %b, cout1 = %b", i1, i2, sum1, cout1);
38
39         $finish;
40     end
41
42     initial begin
43         $dumpfile("full_adder_vector.vcd");
44         $dumpvars(0, full_adder_vector_tb);
45     end
46
47     initial begin
48         $monitor("Time=%0d, i1=%b, i2=%b, sum1=%b, cout1=%b", $time, i1, i2, sum1, cout1);
49     end
50 endmodule

```

```

C:\week4\favector>vvp test
VCD info: dumpfile full_adder_vector.vcd opened for output.
Time=0, i1=000, i2=0, sum1=00, cout1=0
i1 = 000, i2 = 0, sum1 = 00, cout1 = 0
Time=10, i1=001, i2=1, sum1=10, cout1=0
i1 = 001, i2 = 1, sum1 = 10, cout1 = 0
Time=20, i1=011, i2=0, sum1=11, cout1=0
i1 = 011, i2 = 0, sum1 = 11, cout1 = 0
Time=30, i1=111, i2=1, sum1=00, cout1=1
i1 = 111, i2 = 1, sum1 = 00, cout1 = 1
full_adder_vector_tb.v:39: $finish called at 40000 (1ps)

```

